

Markov and Hidden Markov Models

COMP9418 — Advanced Topics in Statistical Machine Learning

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Acknowledgments

- [Barber, BRML, 2012] Bayesian Reasoning and Machine Learning, David Barber, 2012
www.cs.ucl.ac.uk/staff/D.Barber/brml
- [Murphy, MLaPP, 2012] Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012

Dealing with Sequential Data

- So far we have assumed iid data
 - ▶ Likelihood factorizes across observations

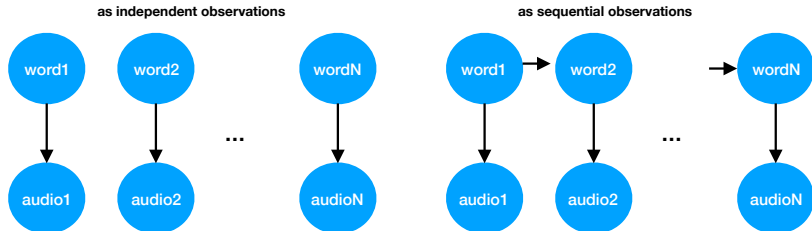
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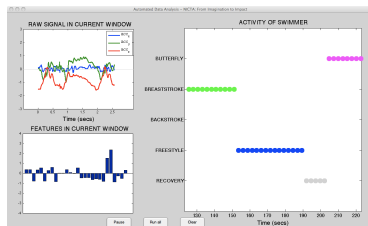
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- This is unrealistic for many situations where data is inherently **sequential**
- Speech recognition: Given an audio waveform, would like to robustly extract & recognize any spoken words



Dealing with Sequential Data

Activity Recognition using Wearable Sensors



Observations may be correlated in time.

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- Temporal data: financial forecasting, currency exchange rate, speech, sensor data, tracking
- Non-temporal data: sequence of characters in an english sentence, sequence of nucleotides in DNA

Aims (1)

This lecture will allow you to understand and apply some common probabilistic models for sequential data. In particular, following it you should be able to:

- Apply fully-observable Markov models to the analysis of sequential data.
- Carry out parameter estimation in first-order Markov chains and understand the complexity of this task for higher-order models.
- Carry out clustering of sequential data via mixtures of Markov chains.
- Understand how the introduction of latent variables captures long-term dependencies.

Aims (2)

- Distinguish the main inference problems, namely smoothing, filtering and prediction, that we can address with Hidden Markov Models (HMMs).
- Solve the above inferential problems along with the most-likely-hidden-path problem via efficient recursions in HMMs.
- Train HMMs for whole-sequence classification and time-dependent classification using a generative or a discriminative approach.

- 1 Markov Chains
- 2 Mixtures of Markov Chains
- 3 Hidden Markov Models

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2 Mixtures of Markov Chains

3 Hidden Markov Models

General Formulation for Fully Observable Models

Notation: We will denote $\mathbf{y}_{1:T} \stackrel{\text{def}}{=} \mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T$

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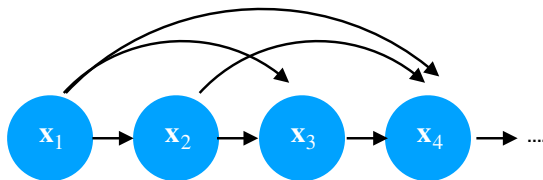
$$p(\mathbf{x}_{1:T}) = p(\mathbf{x}_1) \prod_{t=2}^T p(\mathbf{x}_t|\mathbf{x}_{1:t-1})$$

From General Formulation to Markov Chains

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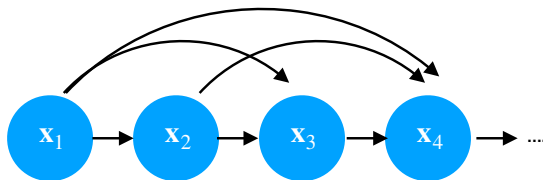
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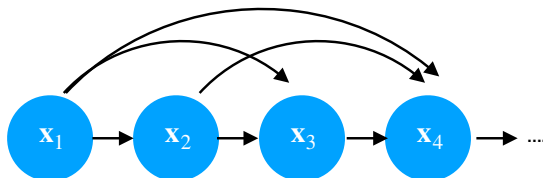
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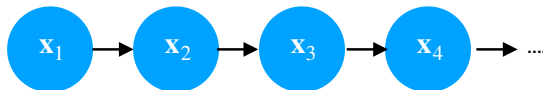
Our initial approach will be to drop some of the long-term dependencies in: $p(\mathbf{x}_t | \mathbf{x}_{1:t-1}) = p(\mathbf{x}_t | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{t-1}) \Rightarrow p(\mathbf{x}_t | \mathbf{x}_{t-1})$

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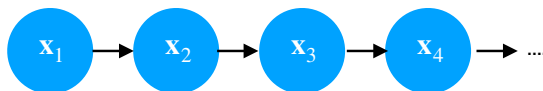


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First Order Markov Chain

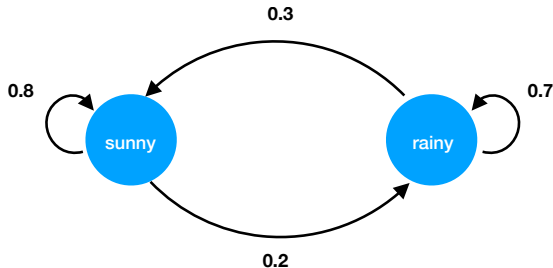
- A Markov chain is defined on either discrete or continuous variables.
- In a first order Markov chain each observation only depends on its immediate past:



$$p(\mathbf{x}_{1:t}) = p(\mathbf{x}_1) \prod_{t=1}^T p(\mathbf{x}_t | \mathbf{x}_{t-1}).$$

Transition Diagram

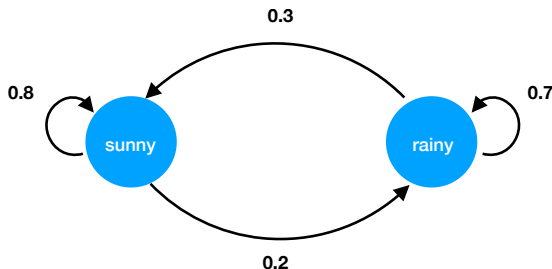
- Discrete variable example:



$$\mathbf{x}_t \in \{\text{sunny}, \text{rainy}\}$$

Transition Diagram

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- If the chain is **stationary** $p(\mathbf{x}_t = \mathbf{s} | \mathbf{x}_{t-1} = \mathbf{s}') = f(\mathbf{s}, \mathbf{s}')$
 - Sometimes this is also called homogeneous

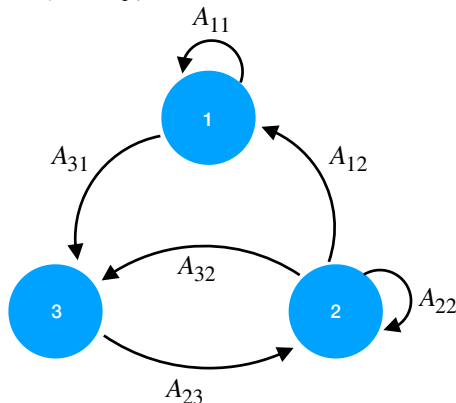
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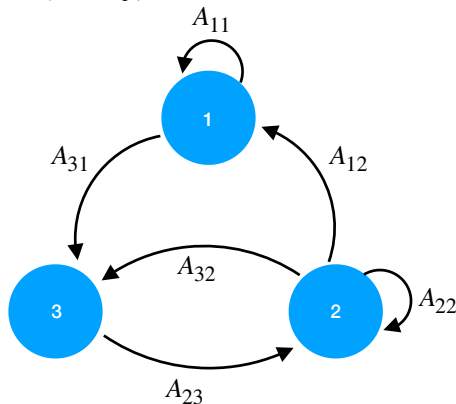
$$\mathbf{A} = \begin{bmatrix} A_{11}, & A_{12}, & 0 \\ 0, & A_{22}, & A_{23} \\ A_{31}, & A_{32}, & 0 \end{bmatrix}$$



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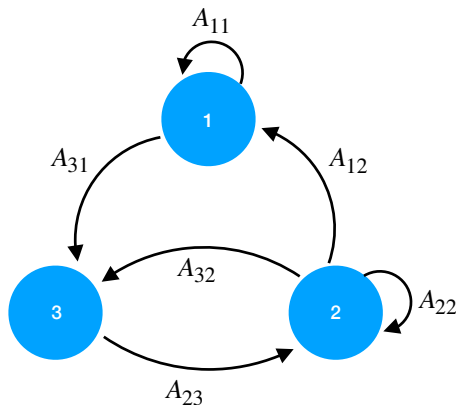
Missing link from i to j simply indicates that $A_{ji} = 0$.

Transition and initial probabilities

- Markov chain:

- ▶ $\mathbf{A}_{K \times K}$: Transition probabilities
- ▶ $\mathbf{p}_1$: Probability of starting the chain in each state at the beginning
- ▶ $\mathbf{x}_t$ can take one of K different states.

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & 0 \\ 0 & A_{22} & A_{23} \\ A_{31} & A_{32} & 0 \end{bmatrix}$$
$$\mathbf{p}_1 = [0.2, 0.4, 0.4]^T$$



Inference on a Markov Chain

Marginal Distribution

Given a discrete-state first-order Markov chain with K states, we are interested in the marginal:

$$p(x_t = i) = \sum_{j=1}^K \underbrace{p(x_t = i | x_{t-1} = j)}_{A_{ij}} p(x_{t-1} = j)$$

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$$\begin{bmatrix} p(x_t = 1) \\ p(x_t = 2) \\ \dots \\ p(x_t = K) \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1K} \\ A_{21} & A_{22} & \dots & A_{2K} \\ \dots & \dots & \dots & \dots \\ A_{K1} & A_{K2} & \dots & A_{KK} \end{bmatrix} \begin{bmatrix} p(x_{t-1} = 1) \\ p(x_{t-1} = 2) \\ \dots \\ p(x_{t-1} = K) \end{bmatrix}$$

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We defined: $\mathbf{p}_t = (p(x_t = 1), p(x_t = 2), \dots, p(x_t = K))^T$.

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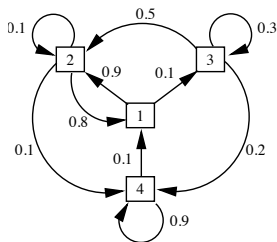
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Then:

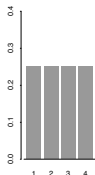
$$\mathbf{p}_2 = \mathbf{A}\mathbf{p}_1, \quad \mathbf{p}_3 = \mathbf{A}\mathbf{p}_2 = \mathbf{A}^2\mathbf{p}_1, \quad \dots \quad \mathbf{p}_t = \mathbf{A}^{t-1}\mathbf{p}_1$$

Equilibrium Distribution of a Markov Chain

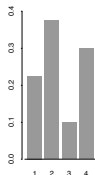
$$\mathbf{p}_1 = [0.25, 0.25, 0.25, 0.25]$$



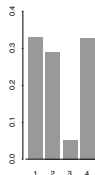
$\mathbf{p}_1$



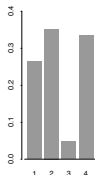
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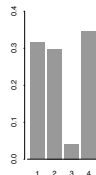
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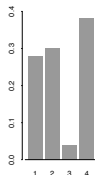
$\mathbf{p}_4$



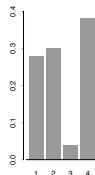
$\mathbf{p}_5$



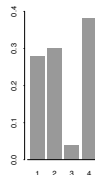
$\mathbf{p}_{501}$



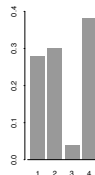
$\mathbf{p}_{502}$



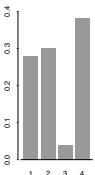
$\mathbf{p}_{503}$



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<https://www.stat.auckland.ac.nz/~fewster/325/notes/ch9.pdf>

Equilibrium Distribution of a Markov Chain

At the equilibrium the marginal distributions does not change anymore:

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For example:

$$\begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{bmatrix} = \begin{bmatrix} 0.0 & 0.8 & 0.0 & 0.1 \\ 0.9 & 0.1 & 0.5 & 0.0 \\ 0.1 & 0.0 & 0.3 & 0.0 \\ 0.0 & 0.1 & 0.2 & 0.9 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{bmatrix}$$

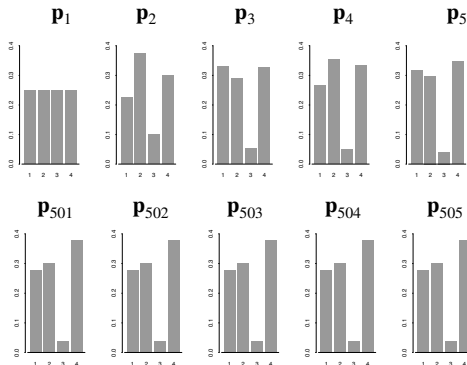
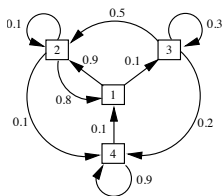
Therefore:

$$[\pi_1, \pi_2, \pi_3, \pi_4]^T = [0.28, 0.30, 0.04, 0.38]^T$$

Equilibrium Distribution of a Markov Chain

- Does equilibrium depend on the starting probabilities?

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- As $t \rightarrow \infty$:

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- Do all Markov chains have an equilibrium distribution?

Equilibrium Distribution of a Markov Chain

PageRank Example (From Barber, BRML, 2012)

Let us define the matrix $\mathbf{H}$ such that:

$H_{ij} = 1$ if website j hyperlinks website i and $H_{ij} = 0$ otherwise.

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An overly simplistic search engine:

- 1 For each website list the words associated with it
- 2 Make an “inverse” list of websites containing word w
- 3 Rank websites containing w according to equilibrium distribution

Learning the Parameters of a Markov Chain

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As usual, we write down the data likelihood:

$$p(\mathcal{D}|\theta) = \prod_{n=1}^N \prod_{i=1}^K (\theta_0^i)^{\mathbb{I}[x_1^n=i]} \prod_{t=2}^T \prod_{j=1}^K (\theta_j^i)^{\mathbb{I}[x_t^n=i, x_{t-1}^n=j]}.$$

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Taking the log and optimizing wrt θ subject to $\sum_{i=1}^K \theta_j^i = 1$:

$$\theta_0^i = \frac{\sum_{n=1}^N \mathbb{I}[x_1^n = i]}{N} \quad \theta_j^i = \frac{\sum_{n=1}^N \sum_{t=2}^T \mathbb{I}[x_t^n = i, x_{t-1}^n = j]}{\sum_{n=1}^N \sum_{t=2}^T \mathbb{I}[x_{t-1}^n = j]}$$

Learning the Parameters of a Markov Chain

Given a set of observation sequences $\mathcal{D} = \{x_{1:T}^n\}_{n=1}^N$ where $x_t \in \{1, \dots, K\}$:

Goal Learn $\theta_0^i \stackrel{\text{def}}{=} p(x_1 = i)$ and $\theta_j^i \stackrel{\text{def}}{=} p(x_t = i | x_{t-1} = j)$ for $t > 1$.

As usual, we write down the data likelihood:

$$p(\mathcal{D}|\theta) = \prod_{n=1}^N \prod_{i=1}^K (\theta_0^i)^{\mathbb{I}[x_1^n=i]} \prod_{t=2}^T \prod_{j=1}^K (\theta_j^i)^{\mathbb{I}[x_t^n=i, x_{t-1}^n=j]}.$$

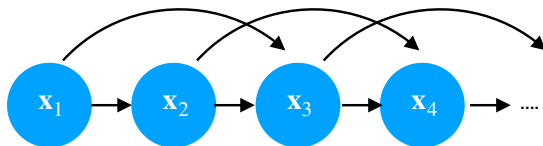
Taking the log and optimizing wrt θ subject to $\sum_{i=1}^K \theta_j^i = 1$:

$$\theta_0^i = \frac{\sum_{n=1}^N \mathbb{I}[x_1^n = i]}{N} \quad \theta_j^i = \frac{\sum_{n=1}^N \sum_{t=2}^T \mathbb{I}[x_t^n = i, x_{t-1}^n = j]}{\sum_{n=1}^N \sum_{t=2}^T \mathbb{I}[x_{t-1}^n = j]}$$

Simply counting occurrences and transitions!

Second Order Markov Chains

We can consider more complex dependencies:



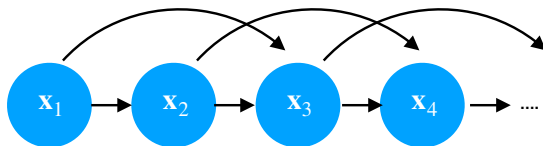
Now the current observation depends on the two previous time steps.

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$$p(\mathbf{x}_t | \mathbf{x}_{t-2}, \mathbf{x}_{t-1}),$$

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The parameterization for the transitions would be:

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which for K -state discrete variables would correspond to $(K - 1)(K^2)$ parameters.

How many parameters did we need for the first order Markov chain?

Higher Order Markov Models

- We can increase the flexibility of our models by considering M^{th} order Markov chains.

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- The number of parameters grows exponentially with the order of the chain
- What can we do?
 - ▶ We need to introduce long-term dependencies while avoiding an explosion in the number of parameters
 - ▶ Our old trick of introducing latent variables
 - ▶ As e.g. in the mixture of Gaussians case

1 Markov Chains

2 Mixtures of Markov Chains

3 Hidden Markov Models

Mixtures of Markov Chains

Example (from BRML 2012):

```
CATAGGCATTCTATGTGCTG
GTGCCTGGACCTGAAAAGCC
GTTGGTCAGCACACGGAAGT
TAAGTGTCTCTGCTCCTAA
GCCAAGCAGGGTCTCAACTT
```

```
CCAGTTACGGACGCCGAAAG
CGGCCGCGCCTCCGGGAACG
CCTCCCCTCCCCTTTCCTGC
CACCATCACCTTGCTAAGG
CATGGACTGCTCCACAAAGG
```

```
TGGAACCTTAAAAAAAAAAAA
AAAGTGCTCTGAAAACTCAC
CACTACGGCTACCTGGGCAA
AAAGAACTCCCCTCCCTGCC
AAAAAACGAAAAACCTAAG
```

```
GTCTCCTGCCCTCTCTGAAC
ACATGAACTACATAGTATAA
CGGTCCGTCCGAGGCACTC
CAAAATGCCTCACGCGTCTCA
GCGTAAAAAAAGTCTGGGT
```

A set of fictitious gene sequences. $\mathbf{x}_t \in \{A, B, C, D\}$, $T = 20$.

Mixtures of Markov Chains

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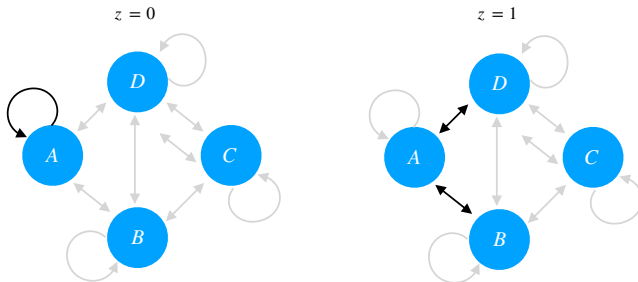
```
CATAGGCATTCTATGTGCTG
GTGCCTGGACCTGAAAAGCC
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```

```
CCAGTTACGGACGCCGAAAG
CGGCCGCGCCTCCGGGAACG
CCTCCCCTCCCCTTTCTGTC
CACCATCACCTTGCTAAGG
CATGGACTGCTCCACAAAGG
```

```
TGGAACCTTAAAAAAAAAAAA
AAAGTGCTCTGAAAACCTCAC
CACTACGGCTACCTGGGCAA
AAAGAACTCCCCTCCCTGCC
AAAAAACGAAAAACCTAAG
```

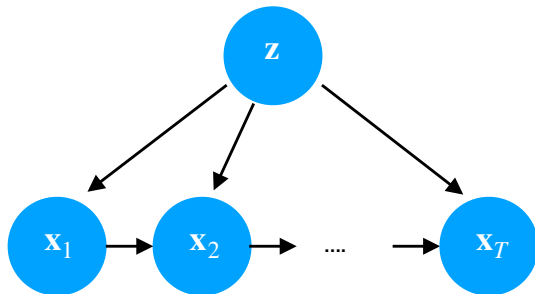
```
GTCTCCTGCCCTCTCTGAAC
ACATGAACTACATAGTATAA
CGGTCCGTCAGGCACTC
CAATGCCTCACGCGTCTCA
CGCTAAAAAAGTCTGGGT
```

A set of fictitious gene sequences. $\mathbf{x}_t \in \{A, B, C, D\}$, $T = 20$. Transition probabilities and starting probabilities are conditioned on the latent variable ($\mathbf{z}$).



Mixtures of Markov Chains

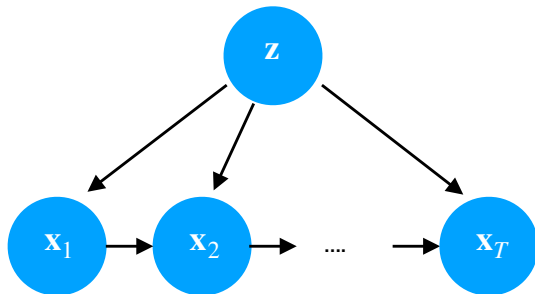
A set of observation sequences $\mathcal{D} = \{x_{1:T}^n\}_{n=1}^N$ where $x_t \in \{1, \dots, K\}$?



- Mixture model with latent variable $z \in \{1, \dots, M\}$.

Mixtures of Markov Chains

A set of observation sequences $\mathcal{D} = \{x_{1:T}^n\}_{n=1}^N$ where $x_t \in \{1, \dots, K\}$?



- Mixture model with latent variable $z \in \{1, \dots, M\}$.
- Markov chain conditioned on the latent variable z :

$$p(x_{1:T}|z) = p(x_1|z) \prod_{t=2}^T p(x_t|x_{t-1}, z)$$

Mixture of Markov Chains

Clustering

- Estimating the parameters of a Mixture Markov Chain:
 - ▶ $p(x_1|z)$
 - ▶ $p(x_t|x_{t-1}, z)$
 - ▶ $p(z)$

Mixture of Markov Chains

Clustering

- Estimating the parameters of a Mixture Markov Chain:
 - ▶ $p(x_1|z)$
 - ▶ $p(x_t|x_{t-1}, z)$
 - ▶ $p(z)$
- Evaluating $p(z|x_{1:T})$:

$$p(z|x_{1:T}) = \frac{p(x_{1:T}|z)p(z)}{\sum_{m=1}^M p(x_{1:T}|z=m)p(z=m)}$$

Mixture of Markov Chains

Model Parameters

Model parameters:

- $\gamma_m^j \stackrel{\text{def}}{=} p(x_1 = j | z = m)$
- $\psi_m^{j,\ell} \stackrel{\text{def}}{=} p(x_t = j | x_{t-1} = \ell, z = m)$
- $\pi_m \stackrel{\text{def}}{=} p(z = m)$

Let $\theta = \{\pi_m, \gamma_m^j, \psi_m^{j,\ell} | m = 1, \dots, M; j, \ell = 1, \dots, K\}$ be the parameters of the model

Mixture of Markov Chains

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As with the GMM, direct likelihood optimization is hard.

Mixture of Markov Chains

Likelihood Maximization via EM

E-step: $P(z^i = m | \mathbf{x}^i, \theta^{old})$

Mixture of Markov Chains

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M-step:

Mixture of Markov Chains

Likelihood Maximization via EM

E-step: $P(z^i = m | \mathbf{x}^i, \theta^{old})$

M-step:

$$\pi_m = \frac{1}{N} \sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old})$$

$$\gamma_m^j = \frac{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \mathbb{I}[x_1^i = j]}{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old})}$$

$$\psi_m^{j,\ell} = \frac{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \sum_{t=2}^T \mathbb{I}[x_t^i = j, x_{t-1}^i = \ell]}{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \sum_{t=2}^T \mathbb{I}[x_{t-1}^i = \ell]}.$$

Mixture of Markov Chains

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$$\begin{aligned}\pi_m &= \frac{1}{N} \sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \\ \gamma_m^j &= \frac{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \mathbb{I}[x_1^i = j]}{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old})} \\ \psi_m^{j,\ell} &= \frac{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \sum_{t=2}^T \mathbb{I}[x_t^i = j, x_{t-1}^i = \ell]}{\sum_{i=1}^N P(z^i = m | \mathbf{x}^i, \theta^{old}) \sum_{t=2}^T \mathbb{I}[x_{t-1}^i = \ell]}.\end{aligned}$$

These are similar updates to the single Markov chain's but weighted by the posterior $P(z^i = m | \mathbf{x}^i, \theta^{old})$.

1 Markov Chains

2 Mixtures of Markov Chains

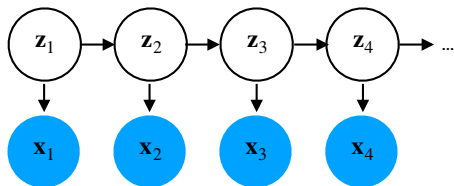
3 Hidden Markov Models

Hidden Markov Models

Introducing Complex Dependencies through Latent Variables

Goal: Efficient ways of modelling long-term dependencies

- ① $\mathbf{z}_t$: latent variable
- ② $\mathbf{x}_t$: observation
- ③ Elements: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$

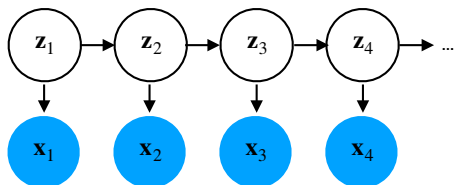


Hidden Markov Models

Examples [from MLaPP, 2012]:

- Automatic speech recognition:

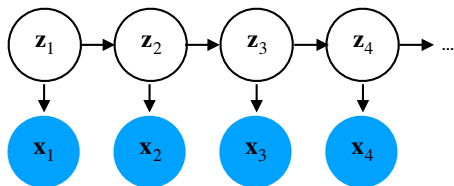
- ▶ $\mathbf{z}_t$: the word being spoken
- ▶ $\mathbf{x}_t$: (acoustic) features extracted speech signal
- ▶ $p(\mathbf{z}_t|\mathbf{z}_{t-1})$: language model
- ▶ $p(\mathbf{x}_t|\mathbf{z}_t)$: acoustic model



Hidden Markov Models

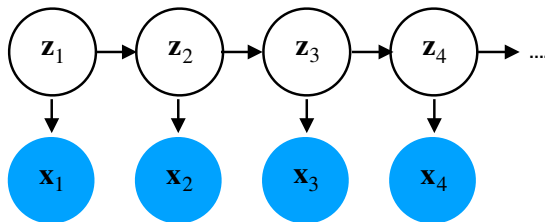
Examples [from MLaPP, 2012]:

- Activity recognition:
 - ▶ z_t : class of activity (running, walking, etc)
 - ▶ x_t : features extracted from video frame
- Part of speech tagging:
 - ▶ z_t : part of speech (noun, verb, adjective, etc.)
 - ▶ x_t : represents a word



Hidden Markov Models

Properties

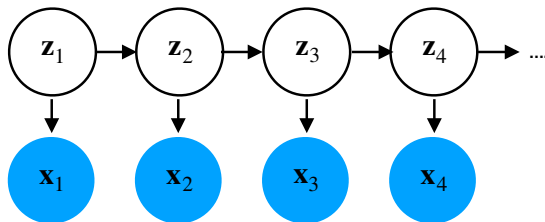


Joint distribution:

$$p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T}) = p(\mathbf{z}_1) p(\mathbf{x}_1 | \mathbf{z}_1) \prod_{t=2}^T p(\mathbf{z}_t | \mathbf{z}_{t-1}) p(\mathbf{x}_t | \mathbf{z}_t)$$

Hidden Markov Models

Properties



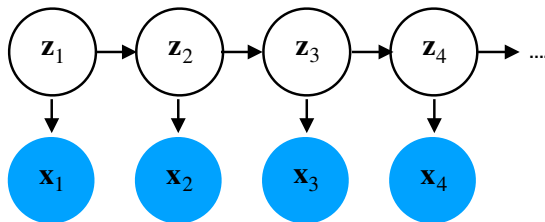
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Predictions for $\mathbf{x}_t$ depend on all the previous observations!

Hidden Markov Models

Properties



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Predictions for $\mathbf{x}_t$ depend on all the previous observations!

Hidden Markov models: latent variables are discrete

Linear dynamical systems: latent and visible variables are Gaussian

Hidden Markov Models

Definitions

Transition Distribution: Assuming $z_t \in \{1, \dots, K\}$ then:

$$A_{ij} = p(z_t = i | z_{t-1} = j)$$

and an **initial distribution**: $\pi_i = p(z_1 = i)$. So we have a table of $K \times K$ probabilities.

Hidden Markov Models

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Emission Distribution:

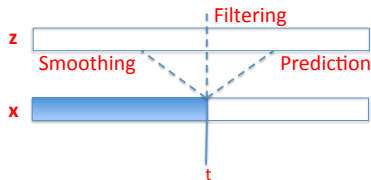
- Discrete states $x_t \in \{1, 2, \dots, S\}$: We have the $S \times K$ emission matrix:

$$B_{ij} = p(x_t = i | z_t = j)$$

- Continuous x_t : z_t selects one of K possible distributions $p(x_t | z_t)$, e.g. a Gaussian: $p(x_t | z_t) = \mathcal{N}(\mathbf{x}_t | \boldsymbol{\mu}_{z_t}, \boldsymbol{\Sigma}_{z_t})$

Hidden Markov Models

Classical Inference Problems

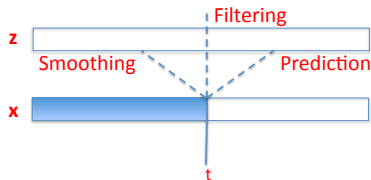


- Filtering: Inferring the present $p(z_t|x_{1:t})$
- Smoothing: Inferring the past: $p(z_u|x_{1:t}), u < t$
- Prediction: Inferring the future: $p(z_s|x_{1:t}), s > t$

- Likelihood: $p(x_{1:T})$
- Most likely hidden path (Viterbi alignment): $\operatorname{argmax}_{z_{1:T}} p(z_{1:T}|x_{1:T})$

Hidden Markov Models

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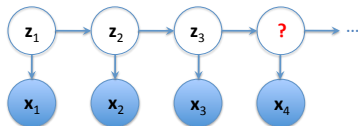
We can use any standard inference method in *graphical models* to solve this problems, e.g. using the Junction Tree Algorithm.

- Instead, we will derive recursions directly.

Hidden Markov Models

Filtering (1)

- Model: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$, $p(\mathbf{z}_1)$

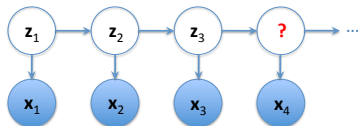


We can find $p(z_t|x_{1:t})$ by considering $p(z_t, x_{1:t})$ and normalizing accordingly.

Hidden Markov Models

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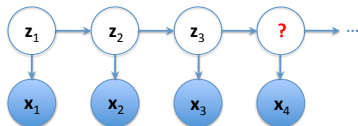
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$$p(z_t, x_{1:t}) = \sum_{z_{t-1}} p(z_t, z_{t-1}, x_{1:t-1}, x_t) \quad \text{Def. marginal prob.}$$

Hidden Markov Models

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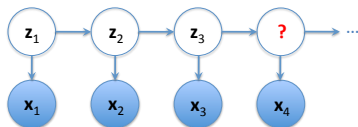
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$$\begin{aligned} p(\mathbf{z}_t, \mathbf{x}_{1:t}) &= \sum_{\mathbf{z}_{t-1}} p(\mathbf{z}_t, \mathbf{z}_{t-1}, \mathbf{x}_{1:t-1}, \mathbf{x}_t) \text{ Def. marginal prob.} \\ &= p(\mathbf{x}_t|\mathbf{z}_t) \sum_{\mathbf{z}_{t-1}} p(\mathbf{z}_t|\mathbf{z}_{t-1})p(\mathbf{z}_{t-1}, \mathbf{x}_{1:t-1}) \text{ used model independencies} \end{aligned}$$

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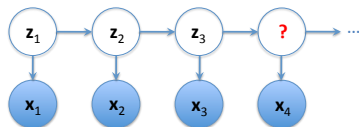
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Hidden Markov Models

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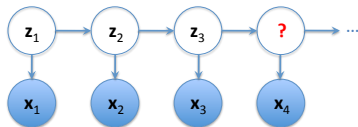
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$$\text{and } \alpha(z_1) = p(z_1)p(x_1|z_1)$$

Hidden Markov Models

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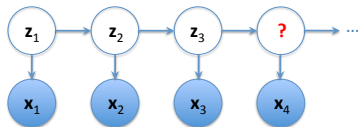
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Hidden Markov Models

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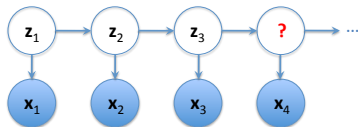
$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1}) \alpha(z_{t-1}), \quad t > 1$$

$$\alpha(z_1) = p(z_1)p(x_1|z_1)$$

Hidden Markov Models

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- Model: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$, $p(\mathbf{z}_1)$



$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1}) \alpha(z_{t-1}), \quad t > 1$$

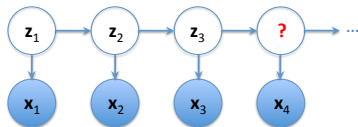
$$\alpha(z_1) = p(z_1)p(x_1|z_1)$$

Example:

Hidden Markov Models

Filtering (1)

- Model: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$, $p(\mathbf{z}_1)$



$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1}) \alpha(z_{t-1}), \quad t > 1$$

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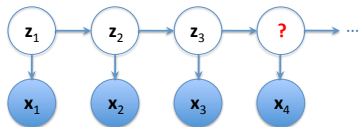
Example:

$$\alpha(z_1 = 1) \dots \alpha(z_1 = M)$$

Hidden Markov Models

Filtering (1)

- Model: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$, $p(\mathbf{z}_1)$



$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1}) \alpha(z_{t-1}), \quad t > 1$$

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Example:

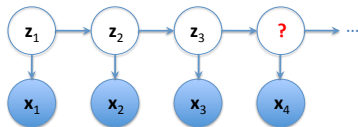
$$\alpha(z_1 = 1) \dots \alpha(z_1 = M)$$

$$\alpha(z_2 = 1) \dots \alpha(z_2 = M)$$

Hidden Markov Models

Filtering (1)

- Model: $p(\mathbf{z}_t|\mathbf{z}_{t-1})$, $p(\mathbf{x}_t|\mathbf{z}_t)$, $p(\mathbf{z}_1)$



$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1}) \alpha(z_{t-1}), \quad t > 1$$

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Example:

$$\alpha(z_1 = 1) \dots \alpha(z_1 = M)$$

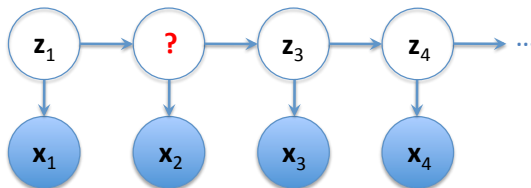
$$\alpha(z_2 = 1) \dots \alpha(z_2 = M)$$

...

$$\alpha(z_t = 1) \dots \alpha(z_t = M)$$

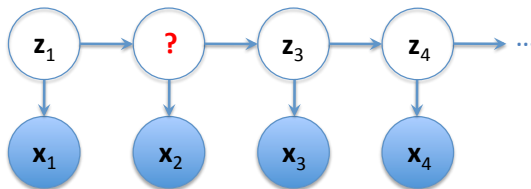
Hidden Markov Models

Smoothing: $p(z_t | x_{1:T})$



Hidden Markov Models

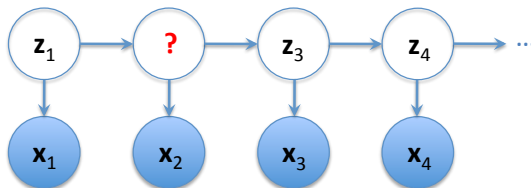
Smoothing: $p(z_t | x_{1:T})$



$$p(z_t, x_{1:T}) = p(z_t, x_{1:t}, x_{t+1:T})$$

Hidden Markov Models

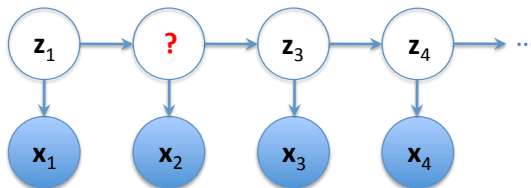
Smoothing: $p(z_t | x_{1:T})$



$$\begin{aligned} p(z_t, x_{1:T}) &= p(z_t, x_{1:t}, x_{t+1:T}) \\ &= p(z_t, x_{1:t}) p(x_{t+1:T} | z_t, x_{1:t}) \quad \text{product rule} \end{aligned}$$

Hidden Markov Models

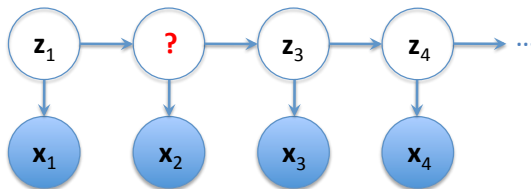
Smoothing: $p(z_t | x_{1:T})$



$$\begin{aligned} p(z_t, x_{1:T}) &= p(z_t, x_{1:t}, x_{t+1:T}) \\ &= p(z_t, x_{1:t}) p(x_{t+1:T} | z_t, x_{1:t}) && \text{product rule} \\ &= \underbrace{p(z_t, x_{1:t})}_{\alpha(z_t)} \underbrace{p(x_{t+1:T} | z_t)}_{\beta(z_t)} && \text{model independencies} \end{aligned}$$

Hidden Markov Models

Smoothing: $p(z_t | x_{1:T})$

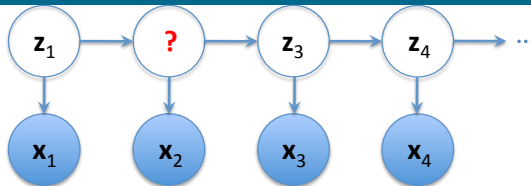


$$\begin{aligned} p(z_t, x_{1:T}) &= p(z_t, x_{1:t}, x_{t+1:T}) \\ &= p(z_t, x_{1:t}) p(x_{t+1:T} | z_t, x_{1:t}) && \text{product rule} \\ &= \underbrace{p(z_t, x_{1:t})}_{\alpha(z_t)} \underbrace{p(x_{t+1:T} | z_t)}_{\beta(z_t)} && \text{model independencies} \end{aligned}$$

Recursive form for $\alpha(z_t)$ as in filtering. **what is $\beta(z_t)$?**

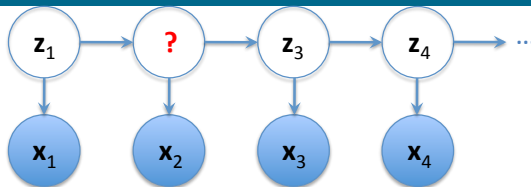
Hidden Markov Models

Smoothing: β recursion



Hidden Markov Models

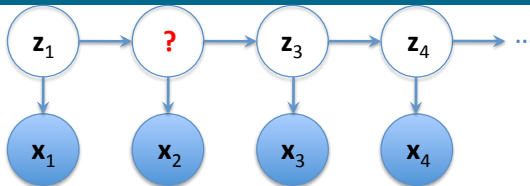
Smoothing: β recursion



$$p(x_{t:T}|z_{t-1}) = \sum_{z_t} p(x_t, x_{t+1:T}, z_t|z_{t-1})$$

Hidden Markov Models

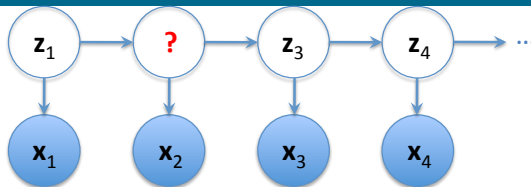
Smoothing: β recursion



$$\begin{aligned} p(x_{t:T}|z_{t-1}) &= \sum_{z_t} p(x_t, x_{t+1:T}, z_t|z_{t-1}) \\ &= \sum_{z_t} p(x_t|z_t) \underbrace{p(x_{t+1:T}|z_t)}_{\beta(z_t)} p(z_t|z_{t-1}) \quad \text{model independencies} \end{aligned}$$

Hidden Markov Models

Smoothing: β recursion

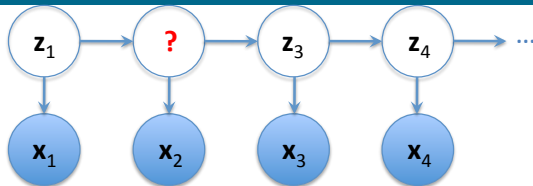


$$\begin{aligned} p(x_{t:T}|z_{t-1}) &= \sum_{z_t} p(x_t, x_{t+1:T}, z_t|z_{t-1}) \\ &= \sum_{z_t} p(x_t|z_t) \underbrace{p(x_{t+1:T}|z_t)}_{\beta(z_t)} p(z_t|z_{t-1}) \quad \text{model independencies} \end{aligned}$$

$$\beta(z_{t-1}) = \sum_z p(x_t|z_t) p(z_t|z_{t-1}) \beta(z_t) \quad \text{backward recursion!}$$

Hidden Markov Models

Smoothing: β recursion



$$\begin{aligned} p(x_{t:T}|z_{t-1}) &= \sum_{z_t} p(x_t, x_{t+1:T}, z_t|z_{t-1}) \\ &= \sum_{z_t} p(x_t|z_t) \underbrace{p(x_{t+1:T}|z_t)}_{\beta(z_t)} p(z_t|z_{t-1}) \quad \text{model independencies} \end{aligned}$$

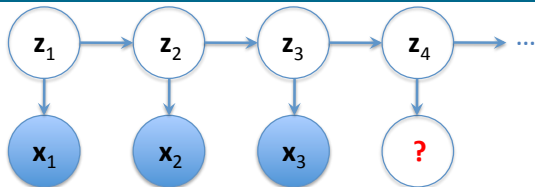
$$\beta(z_{t-1}) = \sum_z p(x_t|z_t) p(z_t|z_{t-1}) \beta(z_t) \quad \text{backward recursion!}$$

$\beta(z_T) = 1$. **forward-backward algorithm** (or $\alpha - \beta$ recursions)

Recursions can be performed in parallel

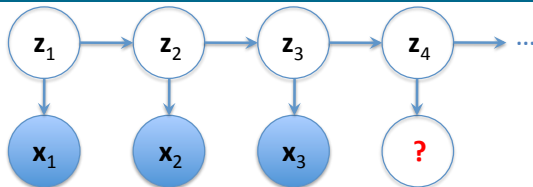
Hidden Markov Models

Prediction and Likelihood



Hidden Markov Models

Prediction and Likelihood

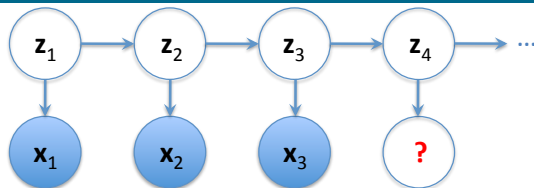


One-step Ahead Prediction:

$$p(x_{t+1}|x_{1:t}) = \sum_{z_t, z_{t+1}} p(x_{t+1}|z_{t+1})p(z_{t+1}|z_t) \underbrace{p(z_t|x_{1:t})}_{\text{filtering}}$$

Hidden Markov Models

Prediction and Likelihood



One-step Ahead Prediction:

$$p(x_{t+1}|x_{1:t}) = \sum_{z_t, z_{t+1}} p(x_{t+1}|z_{t+1})p(z_{t+1}|z_t) \underbrace{p(z_t|x_{1:t})}_{\text{filtering}}$$

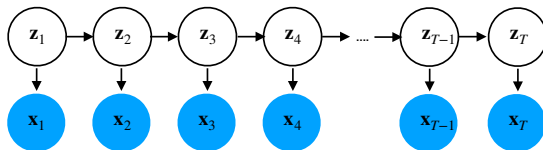
Likelihood Computation:

$$p(x_{1:T}) = \sum_{z_T} p(z_T, x_{1:T}) = \sum_{z_T} \alpha(z_T)$$

It requires only forward computation (filtering).

Hidden Markov Models

Most Likely Hidden Path



We are interested to find $\operatorname{argmax}_{z_{1:T}} p(z_{1:T} | x_{1:T})$.

- We can consider all the possible configurations of $z_{1:T}$ and find the one which has highest $\operatorname{argmax}_{z_{1:T}} p(z_{1:T} | x_{1:T})$.
- But it is computationally expensive $\rightarrow$ **Viterbi Algorithm**

Hidden Markov Models

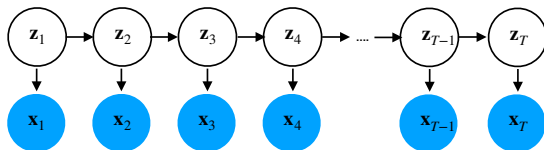
Most Likely Hidden Path

Theorem

if $f(a) \geq 0, \forall a$ and $g(a, b) \geq 0, \forall a, b$ then
$$\max_{a,b} f(a)g(a, b) = \max_a [f(a) \max_b g(a, b)]$$

Hidden Markov Models

Most Likely Hidden Path

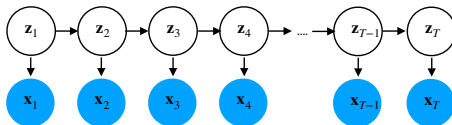


The most likely hidden path of $p(z_{1:T}|x_{1:T})$ is the same as the most likely state of $p(z_{1:T}, x_{1:T})$.

- Consider $\operatorname{argmax}_{z_{1:T}} p(z_{1:T}, x_{1:T})$ instead of $\operatorname{argmax}_{z_{1:T}} p(z_{1:T}|x_{1:T})$

Hidden Markov Models

Most Likely Hidden Path

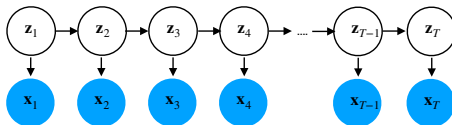


Consider maximizing over $z_{1:T-1}$:

$$\mu_T(z_T) = \max_{z_1 \dots z_{T-1}} p(x_{1:T}, z_{1:T})$$

Hidden Markov Models

Most Likely Hidden Path

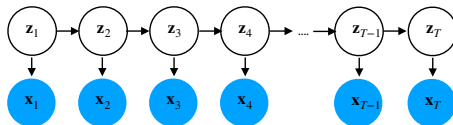


Consider maximizing over $z_{1:T-1}$:

$$\begin{aligned}\mu_T(z_T) &= \max_{z_1 \dots z_{T-1}} p(x_{1:T}, z_{1:T}) \\ &= \max_{z_1 \dots z_{T-1}} \underbrace{p(x_T | z_T) p(z_T | z_{T-1})}_{f(z_{T-1})} \underbrace{\prod_{t=1}^{T-1} p(x_t | z_t) p(z_t | z_{t-1})}_{g(z_{T-1}, z_{1:T-2})}\end{aligned}$$

Hidden Markov Models

Most Likely Hidden Path



Consider maximizing over $z_{1:T-1}$:

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Hidden Markov Models

Most Likely Hidden Path

We defined:

$$\mu_T(z_T) = \max_{z_1 \dots z_{T-1}} p(x_{1:T}, z_{1:T})$$

We showed:

$$\mu_T(z_T) = \max_{z_{T-1}} p(x_T|z_T)p(z_T|z_{T-1}) \underbrace{\max_{z_1 \dots z_{T-2}} \prod_{t=1}^{T-1} p(x_t|z_t)p(z_t|z_{t-1})}_{\mu_{T-1}(z_{T-1})}$$

Hidden Markov Models

Most Likely Hidden Path

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Hence we can define:

$$\mu_t(z_t) = \max_{z_{t-1}} p(x_t|z_t)p(z_t|z_{t-1})\mu_{t-1}(z_{t-1}) \text{ with } \mu_1(z_1) = p(x_1|z_1)p(z_1)$$

Hidden Markov Models

Most Likely Hidden Path

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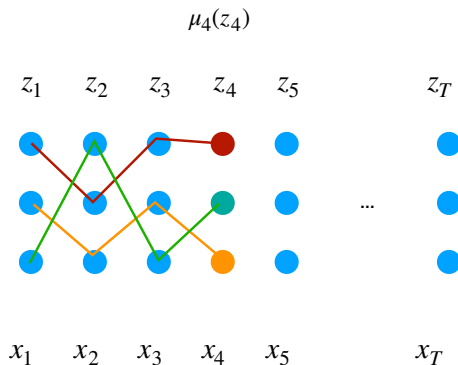
For the path most probably path:

$$\max_{z_T} \mu(z_T)$$

Hidden Markov Models

Most Likely Hidden Path: Example

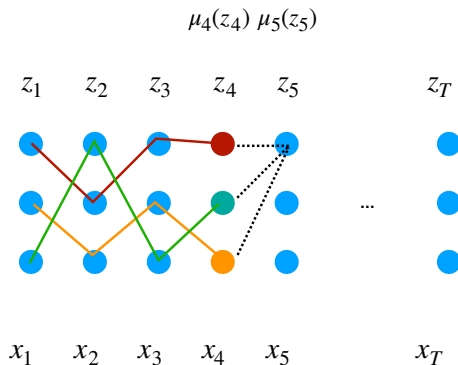
$$\mu_t(z_t) = \max_{z_{t-1}} p(x_t|z_t)p(z_t|z_{t-1})\mu_{t-1}(z_{t-1}) \text{ with } \mu_1(z_1) = p(x_1|z_1)p(z_1)$$



Hidden Markov Models

Most Likely Hidden Path: Example

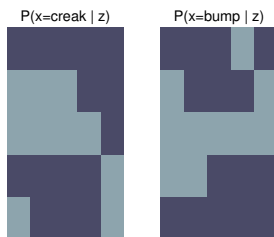
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HMM Classical Inference Problems

Burglar Example (Reproduced from Barber, BRML, 2011)

- You are in bed but have a “mental” partition of the floor as a 5×5 grid
- You know probability of “creak” and “bump” given a position



- Burglar can move only one grid square (left, right, forward, backwards) at time t
- You observe a series of creak/bump information
- Where is the burglar?

HMM Classical Inference Problems

Burglar Example (Reproduced from Barber, BRML, 2011): Setting

- Location of the burglar is hidden. Discrete variable $z \in \{1, \dots, 25\}$
- Absence/presence of creaks and bumps are visible.
- Assume independence and create a new 4-state visible variable using $p(x|z) = p(x^{\text{creak}}|z)p(x^{\text{bump}}|z)$
- How to specify the dynamics?

HMM Classical Inference Problems

Burglar Example (Reproduced from Barber, BRML, 2011): Setting

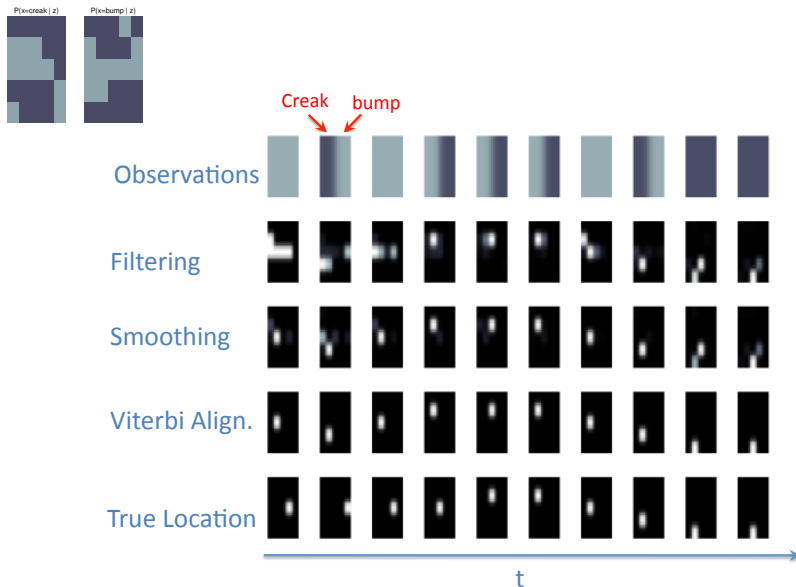
- Location of the burglar is hidden. Discrete variable $z \in \{1, \dots, 25\}$
- Absence/presence of creaks and bumps are visible.
- Assume independence and create a new 4-state visible variable using $p(x|z) = p(x^{\text{creak}}|z)p(x^{\text{bump}}|z)$
- How to specify the dynamics?

Inference questions:

- Where might the burglar be at time t ?
- Where could the burglar have been?
 - ▶ Important info for the police
- Single best guess for sequence of burglar's positions

HMM Classical Inference Problems

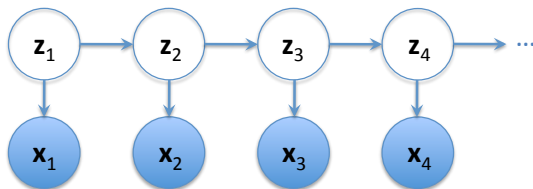
Burglar Example (Reproduced from Barber, BRML, 2011)



Hidden Markov Models

Parameter Learning (1)

Recall our HMM model is given by:



The joint distribution:

$$p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T}) = p(\mathbf{z}_1)p(\mathbf{x}_1|\mathbf{z}_1) \prod_{t=2}^T p(\mathbf{z}_t|\mathbf{z}_{t-1})p(\mathbf{x}_t|\mathbf{z}_t)$$

is parameterized by: $A_{ij} = p(z_t = i | z_{t-1} = j)$, $\pi_i = p(z_1 = i)$ and (assuming discrete observations) $B_{ij} = p(x_t = i | z_t = j)$.

How do we learn these parameters from data?

Hidden Markov Models

Parameter Learning (2)

Given a set of observation sequences $\mathcal{D} = \{x_{1:T}^n\}_{n=1}^N$ where $x_t \in \{1, \dots, S\}$ (assume we know the number of hidden states K)

Goal: learn $\theta = \{\mathbf{A}, \boldsymbol{\pi}, \mathbf{B}\}$

Hidden Markov Models

Parameter Learning (2)

Given a set of observation sequences $\mathcal{D} = \{x_{1:T}^n\}_{n=1}^N$ where $x_t \in \{1, \dots, S\}$ (assume we know the number of hidden states K)

Goal: learn $\theta = \{\mathbf{A}, \boldsymbol{\pi}, \mathbf{B}\}$

We will use EM algorithm.

Hidden Markov Models

EM Algorithm: M-step

In the M-step we need to maximize the objective function:

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \left\langle \sum_{n=1}^N \log p(\mathbf{x}^n, \mathbf{z}^n | \boldsymbol{\theta}) \right\rangle_{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})}$$

Hidden Markov Models

EM Algorithm: M-step

In the M-step we need to maximize the objective function:

$$\begin{aligned} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) &= \left\langle \sum_{n=1}^N \log p(\mathbf{x}^n, \mathbf{z}^n | \boldsymbol{\theta}) \right\rangle_{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})} \\ &= \sum_{n=1}^N \langle \log p(z_1^n | \boldsymbol{\theta}) \rangle_{p(z_1^n | \mathbf{x}_{1:T}^n, \boldsymbol{\theta}^{\text{old}})} \\ &\quad + \sum_{n=1}^N \sum_{t=2}^T \langle \log p(z_t^n | z_{t-1}^n, \boldsymbol{\theta}) \rangle_{p(z_{t-1}^n, z_t^n | \mathbf{x}_{1:T}^n, \boldsymbol{\theta}^{\text{old}})} \\ &\quad + \sum_{n=1}^N \sum_{t=1}^T \langle \log p(x_t^n | z_t^n, \boldsymbol{\theta}) \rangle_{p(z_t^n | \mathbf{x}_{1:T}^n, \boldsymbol{\theta}^{\text{old}})} \end{aligned}$$

Hidden Markov Models

EM Algorithm: M-step

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wrt $\boldsymbol{\theta} = \{\mathbf{A}, \boldsymbol{\pi}, \mathbf{B}\}$ subject to the usual normalization constraints.

Hidden Markov Models

EM Algorithm: M-step: Closed-form updates

Performing the corresponding derivatives we get the following updates:

$$\pi_i = p(z_1 = i) = \frac{1}{N} \sum_{n=1}^N p(z_1^n = i | x_{1:T}^n, \theta^{\text{old}})$$

$$A_{ij} = p(z_t = i | z_{t-1} = j) \propto \sum_{n=1}^N \sum_{t=2}^T p(z_t^n = i, z_{t-1}^n = j | x_{1:T}^n, \theta^{\text{old}})$$

$$B_{sj} = p(x_t = s | z_t = j) \propto \sum_{n=1}^N \sum_{t=1}^T \mathbb{I}[x_t^n = s] p(z_t^n = j | x_{1:T}^n, \theta^{\text{old}})$$

Hidden Markov Models

EM Algorithm: E-step

In the **E-step**, based on the old parameters, we need to update the distributions:

$$p(z_1 = i | x_{1:T}, \theta^{\text{old}}) \quad \text{and} \quad p(z_t = j | x_{1:T}, \theta^{\text{old}})$$

What classical inference problem are we addressing there?

What about $p(z_t = i, z_{t-1} = j | x_{1:T}, \theta^{\text{old}})$?

- This is a pairwise marginal, which can be shown to be:

$$p(z_t, z_{t-1} | x_{1:T}) \propto \alpha(z_{t-1}) p(x_t | z_t) p(z_t | z_{t-1}) \beta(z_t)$$

Hidden Markov Models

EM Algorithm: E-step

In the **E-step**, based on the old parameters, we need to update the distributions:

$$p(z_1 = i | x_{1:T}, \theta^{\text{old}}) \quad \text{and} \quad p(z_t = j | x_{1:T}, \theta^{\text{old}})$$

What classical inference problem are we addressing there?

What about $p(z_t = i, z_{t-1} = j | x_{1:T}, \theta^{\text{old}})$?

- This is a pairwise marginal, which can be shown to be:

$$p(z_t, z_{t-1} | x_{1:T}) \propto \alpha(z_{t-1}) p(x_t | z_t) p(z_t | z_{t-1}) \beta(z_t)$$

This is an interesting example where learning requires the computation of non-straightforward marginals (inference)

Hidden Markov Models

EM Algorithm: Summary

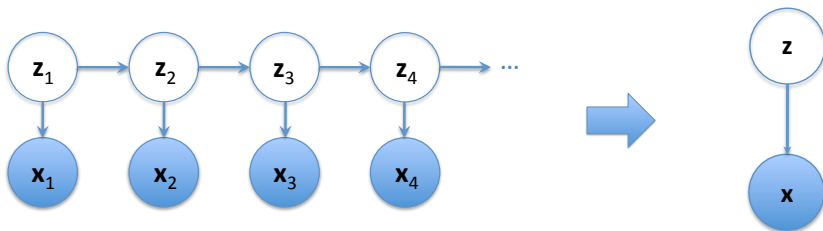
Repeat Until convergence (e.g. using data likelihood):

- 1 Initialize parameters $\theta^{\text{old}} = \{\mathbf{A}, \boldsymbol{\pi}, \mathbf{B}\}$
- 2 Run forward-backward recursions to compute corresponding posteriors $p(z_1 = i | x_{1:T}, \theta^{\text{old}})$, $p(z_t = j | x_{1:T}, \theta^{\text{old}})$, $p(z_t = i, z_{t-1} = j | x_{1:T}, \theta^{\text{old}})$
- 3 Update parameters $\mathbf{A}, \boldsymbol{\pi}, \mathbf{B}$ using these posteriors accordingly
- 4 Evaluate likelihood as convergence criterion

Hidden Markov Models

Parameter Initialization

- EM is plagued with local optima and a good parameter initialization is needed
- This is specially critical for the emission distribution
- Can initialize $p(x_t|z_t)$ with a simple (non-temporal) mixture model, i.e. $p(x) = \sum_z p(z)p(x|z)$



Hidden Markov Models

Continuous Observations

Except for the specific derivation of learning the emission matrix, everything applies to continuous observations.

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- The **HMM-GMM** model uses a mixture of Gaussians as the emission distribution
 - ▶ Popular in tracking and speech recognition

Summary and Conclusions

- Modelling dependencies in sequential data is essential
- Difficult trade-off flexibility vs complexity in fully observable models
- Use trick of introducing latent variables
- Hidden Markov Models are an elegant way of modelling long-term dependencies in observations
- [Reading](#): Barber (BRML, 2017) Ch 23 (except sec 23.4)